

# Single and Multiple Motion Control of Autonomous Robotic Vehicles

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**Abstract**—The past few decades have witnessed a significant research effort in the area of motion control for autonomous robotic vehicles. This paper presents a brief survey related to two topics in this area: *i*) the design of control systems to drive a vehicle to track a time-parameterized trajectory (trajectory tracking), or to follow a geometric path (path-following); and *ii*) the problem of making a group of autonomous robotic vehicles follow desired geometric paths while keeping a specified formation pattern (cooperative path-following). From a theoretical standpoint, special attention is given to a number of challenging problems that include the complex and nonlinear vehicle dynamics, uncertainty, disturbances, and when in cooperation, the problem of communication constraints. Illustrative examples using Lyapunov based techniques for motion control design will be presented for wheeled mobile robots, autonomous underwater vehicles, and unmanned aerial vehicles.

## I. INTRODUCTION

The past few decades have witnessed an increased research effort and an worldwide interest in the use of autonomous robotic vehicles to execute missions of increasing complexity. An attractive and challenging scenario where considerable effort is now being placed is the deployment of groups of heterogeneous networked vehicle systems (air, surface, and underwater) that can interact autonomously with the environment and other vehicles to perform, in the presence of uncertainty and adversity, tasks beyond the ability of individual vehicles. The concept is based on the idea that a monolithic structure can be distributed in an inexpensive network of vehicles, resulting in a significant improvement in efficiency, performance, reconfigurability and robustness, and in the emergence of new capabilities. Some of the potential applications include searching, surveillance and surveying operations, as well as exploration in hazardous environments. It is important to stress that the word autonomous refers to the fact that the robotic vehicles must have sensing and decision capabilities, be aware of their operating conditions and the environment, and be capable of performing tasks without tight human supervision, which clearly introduces core fundamental and applied research problems that need to be solved. These include (to name a few) the presence of nonlinearities, uncertainty, noise and disturbances, the challenge of coupled control/logic communication design, limited and disrupted communication, robustness to hardware malfunctions and failures, complex hierarchical operating objectives, high level coordination/management, etc. These problems pose great challenges to systems engineers, both from a theoretical and

practical standpoint, and many of the simplest ones lies at the boundaries of current tools and understanding. Future progress will require a much deeper understanding of the integration between control, communications, computing, and networks as well as modeling, analysis, and control of complex decision systems.

This paper focus on a particular class of problems mentioned above, and its objective is to provide a brief overview of a number of selected motion control problems for single and multiple autonomous robotic vehicles. We will eschew most of the technical details involved in the specific design and analysis of the control algorithms. Instead, the paper offers a general exposition of the main concepts indicating a comprehensive set of references in the field, together with simple illustrative examples.

In Section II, we start with the description of a simple setup that captures the dynamic model of motion of general autonomous robotic vehicles and then formulates a collection of single motion control problems, which include point stabilization, way-point tracking, trajectory tracking, path following, and moving path following. Then, we proceed with the description of multiple motion control problems. Special attention is given to the challenging task of developing networked control strategies capable of yielding robust performance of a fleet of heterogeneous robotic vehicles in the presence of complex vehicle dynamics, external disturbances, and communication constraints. To this effect, a general architecture is introduced for cooperative path following. In simple terms, the cooperative path following problem, which is at the root of many practical applications, consists of, given  $N$  vehicles and conveniently defined desired paths, designing the motion control algorithms for each vehicle to drive and maintain them in their respective desired paths at a common speed profile and holding a specified formation pattern. Different solutions to this problem and similar problems can be found in the literature, see e.g., [3], [6], [16], [20], [25]–[27], [30], [35]–[38], [42], [54], [58], [61], [65]–[67], [70], and the references there in. An interesting strategy that will be highlighted in this paper consists of decoupling the problem into *i*) a path following problem, where the goal is to derive control laws to drive each vehicle to its path at the reference speed profile, and *ii*) a multiple vehicle coordination problem, where the objective is to adjust the speed of each vehicle so as to achieve the desired formation pattern. The papers [6], [36] offer a theoretical overview of the subject and introduce techniques to solve this problem.

In Section III, we illustrate some of the motion control problems formulated in the previous section using Lyapunov

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based tools for control design. Particular highlighted points include the derivation of motion control laws to solve the trajectory tracking and path following for a mobile robot of the unicycle type, the illustration with simulation results of how a similar line of reason could be applied to solve the trajectory tracking and path following, but now for an underactuated Autonomous Underwater Vehicle (AUV) in 3-D space, and an example for an Unmanned Aerial Vehicle (UAV) that is required to perform a moving path following task, where in the latter we present simulation and experimental results. For the multiple motion control case, we illustrate the cooperative path following applied for three autonomous underwater vehicles in the presence of communication constraints. The paper is closed in Section IV with some conclusions.

## II. MOTION CONTROL PROBLEMS

This section formulates a collection of single and multiple high-level motion control problems for autonomous robotic vehicles. We consider a framework composed by a group of  $N$  autonomous robotic vehicles modeled by general continuous time systems of the form

$$\dot{x}^{[i]}(t) = f^{[i]}(x^{[i]}(t), u^{[i]}(t)), \quad x^{[i]}(0) = x_0^{[i]} \quad (1a)$$

$$y^{[i]}(t) = h^{[i]}(x^{[i]}(t)) \quad (1b)$$

where for each vehicle  $i \in \mathcal{N} := \{1, 2, \dots, N\}$ , the vector  $x^{[i]}(t) \in \mathbb{R}^{n^{[i]}}$  denotes the state,  $x_0^{[i]}$  the initial condition,  $u^{[i]}(t) \in \mathbb{R}^{m^{[i]}}$  the input control signal, and  $y^{[i]}(t) \in \mathbb{R}^{q^{[i]}}$  the output signal that usually corresponds to the position of the vehicle with respect to an inertial coordinate frame.

For simplicity of presentation, we are neglected parametric modeling uncertainty, exogenous disturbances and noise. These are mentioned in the illustrative examples and in the references that will be pointed out along the paper. Also, whenever is clear from the context, we omit the explicit superscript label  $[i]$ . We will now proceed with the definition of some selected motion control problems.

### A. Point stabilization and way-point tracking

This task refers to the problem of steering and stabilizes a vehicle at a given position with a desired orientation. If we relax asymptotic stability (in Lyapunov sense) to just convergence, or even simpler, convergence to just a desired position without orientation constraint, then this task is also refereed to as *Go to Point task*. Notice, however, that convergence do not assure robustness in the sense that the error may not degrade gracefully with the size of the perturbations, [47].

*Definition 1 (Go to point problem):* Consider a single vehicle described by (1) and let  $y_d \in \mathbb{R}^q$  be a given desired point. Design a feedback control law such that all the relevant closed-loop signals are bounded and the regulation error  $\|y(t) - y_d\|$  converges to zero as  $t \rightarrow \infty$ .  $\square$

It is important to stress that the complexity of the point stabilization problem is highly dependent on the configuration of the vehicle under consideration. For nonholonomic

systems, point-stabilization is particularly challenging because as pointed out by Brockett [22], these systems cannot be stabilized by continuously differentiable (or even simply continuous) time invariant static state feedback control laws. To overcome this basic limitation, a variety of approaches have been proposed in the literature. Among the proposed solutions are continuous smooth or almost smooth time-varying (periodic) controllers [29], [39], [51]–[53], [63], [68], discontinuous or piecewise time-invariant smooth control laws [4], [9], [12], [15], [18], [24], and hybrid controllers [5], [19], [40], [49], [59].

Another problem that extends the previous one is that of designing a motion control system to steer the vehicle along a number of selected way-points, until it reaches the final target point, [11].

*Definition 2 (Way-point tracking problem):* Consider a single vehicle described by (1) and let  $p = \{p_1, p_2, \dots, p_n\}$  a given sequence of  $n$  way-points with  $p_i \in \mathbb{R}^q$ ,  $i = 1, 2, \dots, n$ , be a given desired point. Associated with each  $p_i$ , consider the closed ball  $B_{\epsilon_i}$  with center  $p_i$  and radius  $\epsilon > 0$ , i.e.,  $B_{\epsilon_i} := \{p \in \mathbb{R}^q : \|p - p_i\| \leq \epsilon_i\}$ . Design a feedback control law such that all the relevant closed-loop signals are bounded and the signal  $y(t) - p_n$  converges to zero as  $t \rightarrow \infty$  after visiting (that is, reaching) the ordered sequence of neighborhoods  $B_{\epsilon_i}$ ,  $i = 1, 2, \dots, n$ .  $\square$

Way-point tracking can in principle be done in a number of ways. Most of them lend themselves to intuitive interpretation but may lack a solid theoretical background. Perhaps the most widely known is so-called line-of-sight scheme [21], [33], [50]. In this case, vehicle guidance is simply done by issuing heading reference commands to the vehicle's steering system so as to align the main axis of the vehicle along the line of sight between the present position of the vehicle and the way-point to be reached. The tracking of the heading reference command is done via a properly designed autopilot. However, as is well known, separation of guidance and autopilot functions may not yield stability.

### B. Trajectory tracking and path following

Trajectory tracking and path following are typical motion control tasks that autonomous robotic vehicles are required to execute [8], [66]. The trajectory tracking motion control problem is concerned with the design of control laws that force a given vehicle to track a desired trajectory, while in path following, the objective is less restrictive in the sense that it is only required to steer the vehicle towards a path (no time constraints) and make it follow the path with an assigned speed profile that in some cases may be path dependent. In this paper, we adopt the following definitions.

*Definition 3 (Trajectory tracking problem):* Consider a single vehicle described by (1) and let  $y_d(t) : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^q$  be a given desired reference trajectory. Design a feedback control law such that all the relevant closed-loop signals are bounded and the tracking error  $\|y(t) - y_d(t)\|$  converges to zero as  $t \rightarrow \infty$ .  $\square$

*Definition 4 (Path following problem):* Consider a single vehicle described by (1) and let  $y_d(\gamma) : \mathbb{R} \rightarrow \mathbb{R}^q$  be a given desired path parameterized by  $\gamma \in \mathbb{R}$ , and  $v_d \in \mathbb{R}$  a desired speed assignment. Design a feedback control law such that all the relevant closed-loop signals are bounded and the path following error  $\|y(t) - y_d(\gamma(t))\|$  converges to zero as  $t \rightarrow \infty$ , and the parameter  $\gamma$  satisfies the speed assignment  $\dot{\gamma} \rightarrow v_d$  as  $t \rightarrow \infty$ .  $\square$

For wheeled mobile vehicles, the first works on this subject have been presented in a series of groundbreaking papers by Samson et al. (see for example [23] and the references therein). For path following, one classical approach consists of defining the error space using the Serret-Frenet frame concept [17], [46], associated to the path. The same circle of ideas led to the development of trajectory tracking and path following systems for marine vehicles [31], [48] and Unmanned Aerial Vehicles (UAVs) [34], [44], [69]. See also [8], [13] for other approaches not using the Serret-Frenet frame concept.

Notice that in practice, trajectory tracking systems may exhibit lower performance when compared with path following systems and may yield “jerky” maneuvers and large actuator activity. This is because of the temporal constraints imposed, where the vehicle is required to be at a certain point at a desired time. This fact has some considerable implications in terms of performance if the problem setup is a-priori a path following problem. In that case, it is not recommended to implement the naive approach of converting the path following as a tracking problem at the level of the desired reference to track. Indeed, in [1], we investigated limits of performance in reference-tracking and path-following for non-minimum phase systems and highlighted an essential difference between them. While, it is well-known that in the reference-tracking, for non-minimum phase systems, there exists a fundamental performance limitation in terms of a lower bound on the  $L_2$ -norm of the tracking error, even when the control effort is free, we showed that this is not the case for the path following problem, even with the speed assignment imposed. Analog limits of performance for a class of nonlinear systems can be seen in [2], [10]. Despite of path following strategies usually lead to more benign maneuvers, however, there are instances in which one is forced to adopt trajectory tracking strategies; for example, when one wishes to investigate a phenomenon that is strongly time-dependent.

### C. Moving path following

Both in classical trajectory tracking and path following the reference trajectory/path is fixed in space. There are, however, applications where it is useful to follow a path that is attached to a reference frame that moves with respect to an inertial coordinate frame. A typical example is the case of a UAV following a path attached to a vehicle to be tracked, moving with time varying linear and angular velocities with respect to an inertial frame. Note that this problem cannot be directly solved by simply recasting it in a classical path following framework because the target

vehicle imposes time constraints and therefore the expected performance (regarding path following objectives) could be affected. Motivated by this observation, recently in [55]–[57] we present a solution to a new motion control problem, called Moving Path Following (MPF) problem, where the vehicle is required to converge to and follow a desired geometric *moving* path without a specific absolute temporal specification. This method generalizes the classical path following for stationary paths, thus providing a generic tool to follow moving paths that can be applied to several mission scenarios (e.g., terrain/air vehicles tracking [28], [72]) with the advantage of retaining the desirable characteristics of the classical path following method, namely smooth convergence to the moving path and the possibility of doing so at constant speed with respect to an inertial coordinate frame.

*Definition 5 (Moving path following problem):* Consider a single vehicle described by (1) and a given path-transport frame  $\{P\}$  with origin  $p_0$  moving with a desired linear velocity  $v_d = \dot{p}_0$  and angular velocity  $\omega_d$ . Additionally, let  ${}^P y_d(\gamma) : \mathbb{R} \rightarrow \mathbb{R}^q$  be a given desired geometric path expressed in  $\{P\}$ . Design a feedback control law such that all the relevant closed-loop signals are bounded and the moving path following error expressed in the path-transport frame  $\{P\}$ ,  $\|{}^P y(t) - {}^P y_d(\gamma(t))\|$  converges to zero as  $t \rightarrow \infty$ , and the parameter  $\gamma$  satisfies some desired dynamic law  $\dot{\gamma} \rightarrow v_\gamma(\gamma, t)$  as  $t \rightarrow \infty$ .  $\square$

In the above definition, we opted to not introduce explicitly the desired speed assignment (as in the Definition 4) because depending on the application, the desired dynamic law for  $\gamma$  may be implicitly the result of a more convenient specification for the linear velocity of the vehicle, e.g., when we require the vehicle to achieve moving path following with a constant linear forward velocity.

### D. Cooperative path following

In cooperative path following, a fleet of vehicles is tasked to follow a set of desired geometric paths while keeping a desired, possibly time-varying, geometric formation pattern at an assigned formation speed. It will be seen that the implementation of this task calls for the execution of a path following algorithm for each of the vehicles, together with a coordination algorithm that is responsible to change the nominal speeds of the vehicles so as to achieve the desired temporal synchronism. These algorithms should take explicitly into account the topology of the inter-vehicle communications network.

Fig. 1 illustrates a general motion control system architecture for multiple vehicle cooperative motion control. The scheme depicted has been applicable in the scope of several application scenarios. See [6], [36] for a rigorous introduction to the exposed framework using dynamical system concepts where the mathematical tools adopted borrow from graph theory and the theory of input-to-state stable (ISS) systems.

Returning to Fig. 1, the trajectory generator system has the role of providing conveniently desired trajectories/paths for each vehicle according to some specific operation scenario.

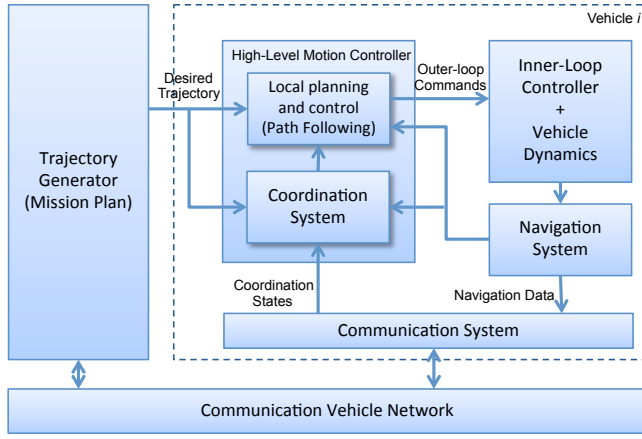


Fig. 1. Multiple vehicle cooperative motion control architecture.

These are computed by taken into account the vehicle roles and their dynamical models, the boundary conditions (e.g., the initial and final poses and velocities of the vehicles), nominal trajectories (usually prescribed as spacial curves not necessarily feasible in terms of the vehicle dynamics), safety constraints (minimal allowed distances to obstacles and other vehicles as well as the map of the location of the obstacles). Depending on the complexity of the application scenario, the trajectory generator usually runs in a base station computer with relatively significant computational power.

The navigation system is in charge of computing the vehicle's state (e.g., linear and rotational positions and velocities of the vehicle) by operating on local navigation data provided by the on-board sensors (e.g., an inertial measurement unit) as well as, when possible, information that is obtained over the communication network. This could be especially relevant in situations where only some of the vehicles can carry accurate navigation suites, whereas the others must rely on less precise sensor suites, complemented with information that is exchanged over the network.

The control and local planning system has the role of generating feasible reference signals (e.g., a generalized collision-free path) according to the mission plan and the vehicle perception, and together with the corresponding navigation data, compute the actuator commands so as to drive suitable tracking errors to zero. In many cases, there exist inner-loop controllers (sometimes denoted as autopilots) that are responsible for sending the low-level command signals to make the vehicle track a given desired signal (e.g., a desired heading) that is provided by the high-level motion controller (that usually operates at the kinematic level).

The coordination system or sometimes called consensus system block is responsible for issuing local correction commands (e.g., speed commands) based on information available over the network so as to achieve the adequate temporal synchronization (and therefore formation) between the vehicles.

Finally, the communication system is responsible for supervising and managing the flow of information (in and

out). Its aim is to establish an interface with the vehicle network through which the vehicles communicate and reduce the frequency at which information is exchanged among the systems involved by incorporating a logic-based communication. This calls for the implementation of systems to decide when and what minimum information should be transmitted from each of the vehicles to its neighbors, see e.g., [6], [43]. Interestingly enough, analogous constraints appear in the vibrant area of networked control systems, from which interesting and fruitful techniques can be borrowed; see for example [41], [71], and the references therein.

In the cooperative path following framework, we impose that the desired geometric paths to be followed for each vehicle are given such that each path is conveniently parameterized by a single variable  $\gamma^{[i]}$ ,  $i \in \mathcal{N}$ , and with the restriction that when the fleet is in the desired formation pattern, the parameterization variables satisfy the equality constraint  $\gamma^{[1]} = \gamma^{[2]} = \dots = \gamma^{[N]}$ . Thus, if each vehicle is endowed with a path-following controller, which means that eventually the vehicles will converge to their desired paths, one can conclude that if we are able to synchronize the path variables  $\gamma^{[i]}$ , which in this context will be renominated as coordination states, it follows that the vehicles will reach the desired formation. To reach consensus the vehicles will have to exchange the coordination states through an inter-vehicle communication network. Typically, all-to-all communications are impossible (or even not desirable) to achieve. The procedure is to achieve coordination by requiring that each vehicle  $i$  will adjust its coordination state speed  $\dot{\gamma}^{[i]}$  as a function only of  $\gamma^{[i]}$  and the values of the neighborhoods coordination states  $\gamma^{[j]}$ ,  $\forall j \in \mathcal{N}^{[i]}$ , where  $\mathcal{N}^{[i]}$  denotes the set of vehicles that vehicle  $i$  communicates with. For simplicity of presentation, throughout this paper, we assume that the communication links are bidirectional, that is  $i \in \mathcal{N}^{[j]} \Leftrightarrow j \in \mathcal{N}^{[i]}$ . A simple way to adjust the speed of the coordination states is to extend the definition of the desired speed profile defined at the path following level by adding a consensus control signal as follows

$$v_d^{[i]} = v_L + u_c^{[i]} \quad (2)$$

where  $v_L$  is the nominal pre-determined formation speed assignment, and the correction speed  $u_c^{[i]}$  is viewed as an input control signal that should vanish when the vehicles are in formation.

We can now state the problem of cooperative path-following.

*Definition 6 (Cooperative path following problem):*

Consider a group of vehicles described by (1) and let  $y_d^{[i]}(\gamma^{[i]}) \in \mathbb{R}^{p^{[i]}}$  be given desired paths parameterized by  $\gamma^{[i]} \in \mathbb{R}$ ,  $i \in \mathcal{I}$ , and  $v_L \in \mathbb{R}$  a desired formation speed assignment. Suppose that the vehicles are supported by an inter-vehicle communication network, and, therefore, for each vehicle  $i \in \mathcal{N}$ , the variables  $\gamma^{[i]}$  and  $\gamma^{[j]}$ ,  $j \in \mathcal{N}^{[i]}$  are available. Design feedback control laws such that all the relevant closed-loop signals are bounded, and as  $t$  approaches to infinity,

- 1) The path following errors converge to zero;

- 2) for each pair of the vehicles  $i, j \in \mathcal{N}$ , the coordination errors  $(\gamma^{[i]} - \gamma^{[j]})$  converge to zero;
- 3) The signals  $\dot{\gamma}^{[i]}(t)$ ,  $i \in \mathcal{N}$  converge to the desired common speed assignment  $v_L$ .

□

### E. Time-critical cooperative path following

Similar to the cooperative path following case, this task addresses the problem of steering a fleet of vehicles along desired geometric paths but now they should meet stringent spatial and temporal constraints. Representative examples include situations where a group of unmanned aerial vehicles are tasked to cooperatively execute collision-free maneuvers and arrive at their final destinations at the same time, or at different times so as to meet a desired inter-vehicle arrival schedule (e.g., sequential auto-landing). Note that in both cases, no absolute temporal constraints are given a priori. In [25], [45], [70], we make use of the cooperative path following framework to solve this type of problems. In particular, in [70], the vehicles are assigned nominal spatial paths and speed profiles along those, and then the vehicles are requested to execute cooperative path following, rather than open loop trajectory tracking maneuvers. This strategy yields robust behavior against external disturbances by allowing the vehicles to negotiate their speeds along the paths in response to information exchanged over the supporting communications network. The work considers the case where the graph that captures the underlying time-varying communications topology is disconnected during some interval of time or even fails to be connected at all times. Conditions were given under which the cooperative path-following closed-loop system is stable. Flight test results of a coordinated road-search mission demonstrated the efficacy of the multi-vehicle cooperative control framework developed in the scope of that work.

## III. ILLUSTRATIVE EXAMPLES USING A LYAPUNOV-BASED APPROACH

This section illustrate some of the motion control problems formulated in the previous section. We will focus only on Lyapunov based control designs. It is important however to stress that there exist other design methods (e.g., linear tools) and more recently optimization based tools that may offer in some cases, some advantages when compared with Lyapunov based methods, see e.g., [7], [13], [14], [62].

### A. Trajectory tracking for a mobile robot of the unicycle type

1) *Model*: Following standard practice, the equations of motion of a rigid body can be developed using an inertial coordinate frame  $\{I\}$  and a body-fixed coordinate frame  $\{B\}$ . For a unicycle type wheeled mobile robot, the kinematic equations take the form

$$\dot{x} = v \cos \theta \quad (3a)$$

$$\dot{y} = v \sin \theta \quad (3b)$$

$$\dot{\theta} = \omega \quad (3c)$$

where  $p := (x(t), y(t))' \in \mathbb{R}^2$  denotes the cartesian coordinates of the vehicle's center of mass,  $\theta(t) \in \mathbb{R}$  its orientation (heading angle), and  $v(t) \in \mathbb{R}$  and  $\omega(t) \in \mathbb{R}$  denote the linear and angular velocities, respectively. The equations in (3) can be written compactly as

$$\dot{p} = Rv \quad (4a)$$

$$\dot{R} = R\Omega(\omega) \quad (4b)$$

with  $\nu(t) := (v(t), 0)' \in \mathbb{R}^2$ . The pair  $(p(t), R(t)) \in SE(2)^1$  is the configuration of the vehicle, position and orientation, where  $R(t) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$  is the rotation matrix from body to inertial coordinates, and  $(v(t), \Omega(\omega(t))) \in se(2)$  is the twist that defines the velocity of the vehicle, linear and angular, where the matrix  $\Omega(\omega(t))$  is the skew-symmetric matrix associated to the angular velocity  $\omega(t) \in \mathbb{R}$ , defined as  $\Omega(\omega) := \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$ . In this example, the control input is defined to be the velocities (linear and angular) according to

$$u(t) := (v(t) \quad \omega(t))' \quad (5)$$

2) *Lyapunov-based controller*: Suppose that we wish to drive the robot, and more precisely, the point

$$y(t) := p(t) + R(t)\epsilon \quad (6)$$

that is placed at a constant distance  $\epsilon = (\epsilon_1, \epsilon_2)' \in \mathbb{R}^2$  from the center of rotation of the body frame  $\{B\}$ , to a desired trajectory  $y_d : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^2$  that is assumed to be continuously differentiable. Then, defining the body-fixed tracking error

$$e(t) := R'(y(t) - y_d(t))$$

and computing its time derivative results in the following error dynamical model

$$\begin{aligned} \dot{e} &= -\Omega R'(y - y_d) + R'(Rv + R\Omega\epsilon - \dot{y}_d) \\ &= -\Omega e + \Delta u - R'\dot{y}_d \end{aligned} \quad (7)$$

with  $\Delta := \begin{pmatrix} 1 & -\epsilon_2 \\ 0 & \epsilon_1 \end{pmatrix}$ . Introducing now the control Lyapunov function  $V := \frac{1}{2}e'e$ , it follows that its time derivative satisfies

$$\dot{V} = e'(-\Omega e + \Delta u - R'\dot{y}_d) = e'(\Delta u - R'\dot{y}_d)$$

where we used the fact that  $\Omega$  is a skew-symmetric matrix and therefore  $e'\Omega e = 0$  for any  $e \in \mathbb{R}^2$ . Thus, imposing  $\epsilon_1$  to be different from zero so that the matrix  $\Delta$  is full rank, it follows that the control input

$$u(t) = \Delta^{-1}(R'\dot{y}_d - Ke) \quad (8)$$

for some design parameter  $K > 0$  makes the origin  $e = 0$  an exponentially stable equilibrium point because

$$\dot{V} = -e'Ke \leq -\|e\|^2 \lambda_{\min}(K) = -2\lambda_{\min}(K)V. \quad (9)$$

Thus, (8) solves the trajectory tracking problem.

<sup>1</sup>For a given  $n \in \mathbb{N}$ ,  $SE(n)$  denotes the Cartesian product of  $\mathbb{R}^n$  with the group  $SO(n)$  of  $n \times n$  rotation matrices and  $se(n)$  denotes the Cartesian product of  $\mathbb{R}^n$  with the space  $so(n)$  of  $n \times n$  skew-symmetric matrices.

### B. Path following for a mobile robot of the unicycle type

Using the derivations discussed in the previous section, we now extend them to solve the path-following problem. Consider (6) and define the body-fixed path following error

$$e(t) := R'(y(t) - y_d(\gamma(t)))$$

Computing its time derivative results

$$\dot{e} = -\Omega e + \Delta u - R' y_d' \dot{\gamma} \quad (10)$$

with  $\Delta$  as before, and  $y_d' := \frac{\partial y_d}{\partial \gamma}$ . Thus a simple path-following controller can be given by the dynamical system

$$\dot{\gamma}(t) = v_d \quad (11a)$$

$$u(t) = \Delta^{-1}(R' y_d' v_d - K e) \quad (11b)$$

Notice that the Lyapunov function  $V := \frac{1}{2} e' e$  satisfies a similar inequality (9), which means that  $y(t)$  converges to the geometric path  $y_d(\gamma)$  and  $\dot{\gamma}(t)$  satisfies (for every time) the speed assignment. Another alternative could be to define a speed assignment error  $\eta := \dot{\gamma} - v_d$ , introduce an augmented Lyapunov function  $V_2 := V + \frac{1}{2} \eta^2$  and define conveniently a control law for  $\dot{\gamma}$  so that  $\dot{V}_2$  is negative definite.

### C. Trajectory tracking and path following for an underactuated AUV in 3-D space

A similar line of reason, but more involved, can be applied for a more general case. We now illustrate one such case.

1) *Model*: In this example, we consider a larger class of underactuated robotic vehicles modeled as a rigid body subject to external forces and torques. Similar to the previous example, let  $\{\mathcal{I}\}$  be an inertial coordinate frame and  $\{\mathcal{B}\}$  a body-fixed coordinate frame whose origin is located at the center of mass of the vehicle. In this case, the configuration  $(R, p)$  of the vehicle is an element of the Special Euclidean group  $SE(3) := SO(3) \times \mathbb{R}^3$ , where  $R \in SO(3) := \{R \in \mathbb{R}^{3 \times 3} : RR' = I_3, \det(R) = +1\}$  is a rotation matrix that describes the orientation of the vehicle and maps body coordinates into inertial coordinates, and  $p \in \mathbb{R}^3$  is the position of the origin of  $\{\mathcal{B}\}$  in  $\{\mathcal{I}\}$ . Denote by  $\nu \in \mathbb{R}^3$  and  $\omega \in \mathbb{R}^3$  the linear and angular velocities of the vehicle relative to  $\{\mathcal{I}\}$  expressed in  $\{\mathcal{B}\}$ , respectively. The following kinematic relations apply

$$\dot{p} = R\nu \quad (12a)$$

$$\dot{R} = R\Omega(\omega) \quad (12b)$$

where  $\Omega(\cdot)$  is a function from  $\mathbb{R}^3$  to the space of skew-symmetric matrices  $\mathcal{S} := \{M \in \mathbb{R}^{3 \times 3} : M = -M'\}$  defined by

$$\Omega(x) := \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}, \quad \forall x := (x_1, x_2, x_3)' \in \mathbb{R}^3.$$

We consider underactuated vehicles with dynamic equations of motion of the following form:

$$M\dot{\nu} = -\Omega(\omega)M\nu + f_\nu(\nu, p, R) + g_\nu u_\nu \quad (13a)$$

$$J\dot{\omega} = -\Omega(\nu)M\nu - \Omega(\omega)J\omega + f_\omega(\nu, \omega, p, R) + G_\omega u_\omega \quad (13b)$$

where  $M \in \mathbb{R}^{3 \times 3}$  and  $J \in \mathbb{R}^{3 \times 3}$  denote constant symmetric positive definite mass and inertia matrices;  $u_\nu(t) \in \mathbb{R}$  and  $u_\omega(t) \in \mathbb{R}^3$  denote the control inputs, which act upon the system through a constant nonzero vector  $g_\nu \in \mathbb{R}^3$  and a constant nonsingular matrix<sup>2</sup>  $G_\omega \in \mathbb{R}^{3 \times 3}$ , respectively; the terms  $-\Omega(\omega)M\nu$  in (13a) and  $-\Omega(\nu)M\nu - \Omega(\omega)J\omega$  in (13b) are the rigid-body Coriolis terms, and the  $C^1$  functions  $f_\nu(\cdot)$ ,  $f_\omega(\cdot)$  represent all the remaining forces and torques acting on the body. For the special case of an underwater vehicle,  $M$  and  $J$  also include the so-called hydrodynamic added-mass  $M_A$  and added-inertia  $J_A$  matrices, respectively, i.e.,  $M = M_{RB} + M_A$ ,  $J = J_{RB} + J_A$ , where  $M_{RB}$  and  $J_{RB}$  are the rigid-body mass and inertia matrices, respectively.

For an underactuated vehicle restricted to moving on a planar surface, the same equations of motion (12)–(13) apply without the first two right-hand-side terms in (13b). Also, in this case,  $(R, p) \in SE(2)$ ,  $\nu \in \mathbb{R}^2$ ,  $\omega \in \mathbb{R}$ ,  $g_\nu \in \mathbb{R}^2$ ,  $G_\omega \in \mathbb{R}$ ,  $u_\omega \in \mathbb{R}$ , with all the other terms in (13) having appropriate dimensions, and the skew-symmetric matrix  $\Omega(\omega)$  be defined as in Section III-A.1. For simplicity, in what follows, we restrict our attention to the three-dimensional case. However, all results are directly applicable to the two-dimensional case.

2) *Simulation results*: In [8], we propose an iterative design to solve the trajectory tracking and path following problem for the model previously described. The methodology adopted is rooted in Lyapunov-based and backstepping techniques and follows a similar line of reason that was presented for the unicycle mobile robot. In this section, we illustrate the behavior of the motion control laws derived in [8] for an ellipsoidal shaped underactuated Autonomous Underwater Vehicle (AUV). We consider that we have available a pure body-fixed control force  $\tau_u$  in the  $x_B$  direction, and two independent control torques  $\tau_q$  and  $\tau_r$  about the  $y_B$  and  $z_B$  axes of the vehicle, respectively. The kinematics and dynamics equations of motion of the vehicle can be written as (12)–(13), see details in [8].

Fig. 2(a) displays the vehicle trajectory using the trajectory-tracking controller in the 3D-space for the following desired trajectory

$$y_d(t) = [V_1 \cos(\frac{2\pi}{T}t + \phi_d), V_1 \sin(\frac{2\pi}{T}t + \phi_d), V_2 t]$$

with  $V_1 = 20$ ,  $V_2 = 0.05$ ,  $T = 400$ , and  $\phi_d = -\frac{\pi}{2}$ . The initial conditions of the AUV are  $p_0 = (x_0, y_0, z_0) = (10\text{ m}, -10\text{ m}, 0)$ ,  $R = I$ , and  $v_0 = \omega_0 = 0$ . Figures 2(c) and 2(e) show the time evolution of the tracking error  $e$  and the Euler angles (computed from  $R$ ), respectively. The damped oscillatory behavior of pitch and roll are due to the gravitational and buoyancy forces. Notice that the initial position of the desired position was deliberately chosen to be almost behind the initial position of the vehicle. As we can see, the vehicle turns back in its attempt to be at the given reference position at the prescribed time, requiring significant control effort and consequently inducing a strong oscillatory behavior.

<sup>2</sup>This can be relaxed, see [8] for the special case of  $G_\omega \in \mathbb{R}^{3 \times 2}$ .

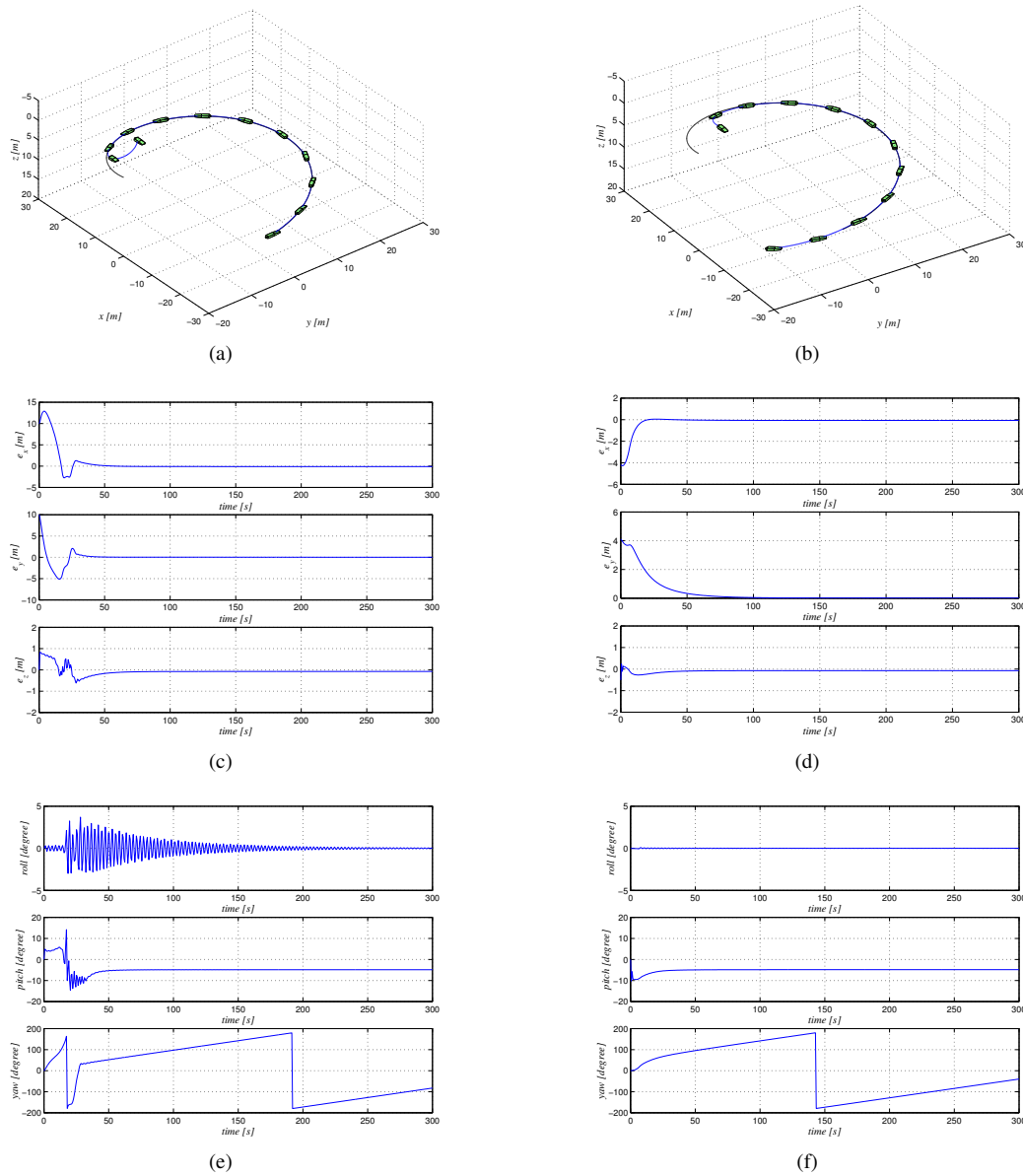


Fig. 2. Vehicle trajectory in 3-D space using a trajectory-tracking controller (Fig. (a)), and the path-following controller (Fig. (b)). Time evolution of the position error  $e = (e_x, e_y, e_z)$ , the roll  $\phi$ , pitch  $\theta$ , and yaw  $\psi$  Euler angles for the trajectory-tracking (Fig. (c), (e)), and the path-following (Fig. (d), (f)).

Figures 2(b), 2(d), and 2(f) show the behavior of the path following controller, where the desired path, the initial conditions and the control gains are the same as in the experiment for the tracking controller, and the speed assignment was set to  $v_d = 1 \text{ m/s}$ . The initial condition for  $\gamma$  was chosen to be the one that minimizes the distance between the initial position of the AUV and the desired path. The convergence of the vehicle to the path is now much smoother. From these two experiments one can see that when the primary objective is to steer the vehicle to converge to and move along a geometric path, the path-following controller offers significant performance improvement. For simplicity, in these experiments we did not include model uncertainty, but [8] explains how the Lyapunov based control results can be extended using an adaptive supervisory control algorithm

that employs a logic-based switching scheme, to address the presence of possibly large modeling parametric uncertainty.

#### D. Moving path following of a fixed-wing UAV

The motivation of this section is twofold: 1) present an alternative error model to what was presented so far for motion control problems by using a Serret-Frenet frame approach; and 2) describe a solution for a simple moving path following problem. The exposition of this section follows directly from the recent results described in [56].

1) *Error model:* For simplicity of presentation, we will focus on the planar case. The general 3-D case can be seen in [57]. Fig. 3 illustrates a moving path following application example where a UAV should follow a planar (horizontal) path moving with linear velocity  $v_d$  and angular velocity



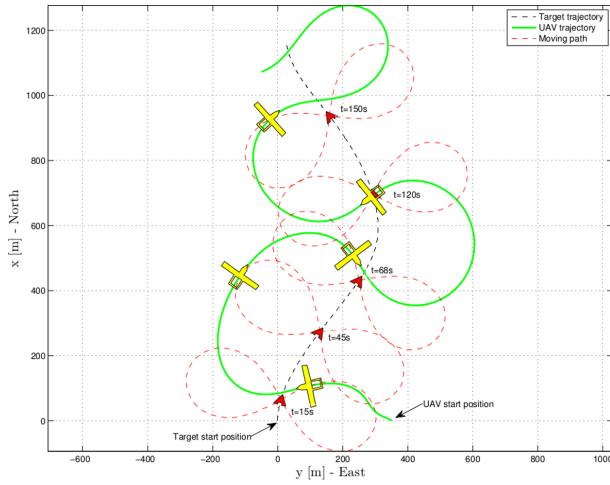


Fig. 4. Numerical simulation - aircraft's trajectory following a target between  $t=0s$  and  $t=200s$ .

where  $g_1$  and  $g_2$  are positive scalars assigning relative weights between position and orientation errors, to be

$$\dot{V}_1 = -\frac{g_1}{g_2} \tilde{\psi}^2 \leq 0. \quad (16)$$

Using standard Lyapunov arguments that makes uses of the Barbalat lemma [47], it can be concluded that the proposed controller solves the moving path following problem.

2) *Simulation results:* Fig. 4 illustrates an application of the MPF control law to a single ground target tracking scenario where a UAV flying at 20m/s airspeed is required to track a target by following a lemniscate path that is moving together with the target with 300m width, keeping the line that connects the two foci always perpendicular to  $\psi_d$  (thus  $\dot{\psi}_p = \dot{\psi}_d$ ). The wind velocity  $W_t$  is set to 10m/s (blowing from South) between 80 and 150 seconds (simulation time) and is set to zero otherwise. The target is moving with time-varying linear and angular velocities according to

$$\begin{aligned} (p_{tx}, p_{ty}, \psi_t, \|v_t\|)|_{t=0} &= (0m, 0m, 0, 4m/s) \\ \|\dot{v}_t\| &= 0.2 \sin(0.07t) m/s^2 \\ \dot{\psi}_t &= 0.02 \cos(0.03t) rad/s \end{aligned} \quad (17)$$

where  $v_t$  corresponds to the target velocity and  $\psi_t$  is the target heading. In order to attach the desired path to the target we set the path fixed frame with  $p_0|_{t=0} = [p_{tx} \ p_{ty}]|_{t=0}$ ,  $v_d = v_t$  and  $\psi_d = \psi_t$ .

Fig. 5 demonstrates the performance of the overall control system. At time instants  $t = 80s$  and  $t = 150s$  a perturbation on the distance and heading errors due to the imposed sudden change on the wind velocity is observed.

3) *Flight tests:* This section presents the field tests for the single target tracking scenario. In this experiment, the goal was to make the UAV track a ground vehicle by following a lemniscate centered at the target position with a 300m width, keeping the line that connects the two foci always perpendicular to  $\psi_t$ . The control algorithm was implemented and tested on the UAV ANTEX-X02 (see [56] for details). The UAV is equipped with a Piccolo II autopilot that plays

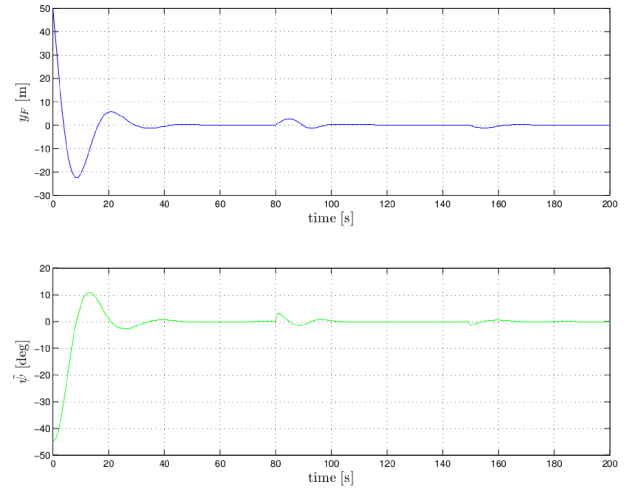


Fig. 5. Numerical simulation - position and heading errors.

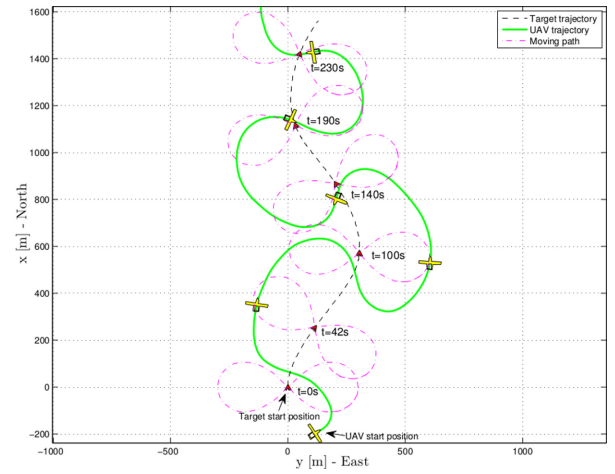


Fig. 6. Aircraft's trajectory following a target on the ground (from  $t = 0s$  to  $t = 250s$ ).

the role of an inner-loop controller that provides the required actuation signals for the UAV's control surface's deflections and the engine power. The UAV was flying at constant altitude - 200m above the ground - with 20m/s commanded airspeed. The wind was blowing from south-east with 3m/s average speed and the controller parameters were set to  $g_1 = 0.22$  and  $g_2 = 0.00012$ . The target trajectory was emulated using the same parameters as in the numerical simulation (cf. equations (17)).

The obtained qualitative behaviour of the UAV is similar to the results presented for the numerical simulation. Fig. 6 shows the overall UAV and target trajectories. The UAV, target and path positions at sample time instants are also presented. Fig. 7 shows that the bank command and the bank value are kept within the linear region. There is a non negligible delay between the commanded bank and its actual value thus decreasing system performance. However, distance and angular errors plots in Fig. 8 demonstrate the good performance of the implemented control system.

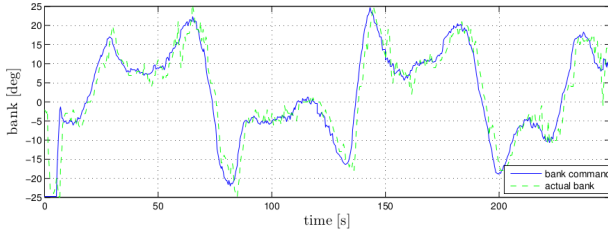


Fig. 7. Commanded and actual bank for the single target tracking scenario.

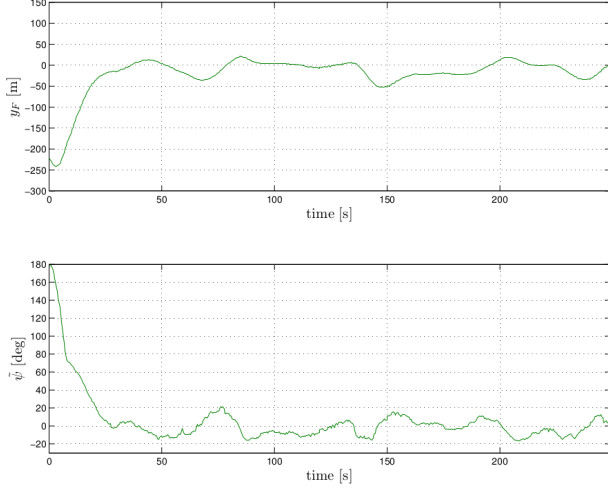


Fig. 8. Distance and heading errors for the single target tracking scenario.

### E. Cooperative path following of a formation of 3 AUVs

This section illustrates the application of the cooperative path following to the formation control of three autonomous underwater vehicles. In this example, the idea is to make use of the path following algorithm presented in Section III-C and as described in Section II-D add in the desired speed profile a consensus control signal  $u_c$ , see equation (2). Note that one have to guarantee that the path following closed-loop system is input-to-state stable (ISS) from  $u_c$  as an input. A second stage is to design the coordination controller that has the responsibility to output a correction speed signal  $u_c$  to adjust the speed of the vehicle so as to synchronize its position with the other vehicles and achieve vehicle coordination.

1) *Coordination*: From the adopted cooperative path following setup, to achieve the specified formation, the vehicles only have to exchange the coordination states (which are the path-parameterizations) over the supported communication network to reach agreement on these states. From the theory of consensus of distributed systems and assuming that each vehicle receives information from its neighbors continuously, a natural choice for the coordination control law would be the so-called neighboring rule, see, e.g., [6]

$$u_c^{[i]} = -\kappa \sum_{j \in \mathcal{N}^{[i]}} (\gamma^{[i]} - \gamma^{[j]}), \quad (18)$$

where  $\kappa$  is a positive scalar. To reduce the communication rate and lift the continuously transmissions, we could con-

sider instead that each vehicle relies on their estimates that are updated at the transmission times  $t = t_k$ . In this case, one solution is the following modification

$$u_c^{[i]} = -\kappa \sum_{j \in \mathcal{N}^{[i]}} (\hat{\gamma}_i^{[i]} - \hat{\gamma}_j^{[i]}), \quad (19)$$

where  $\hat{\gamma}_\gamma^{[i]}$  is an estimate of  $\gamma^{[\gamma]}$  running on agent  $i$ . The estimators of the coordination states in each UAV run open-loop most of the time but are reset once new data is received from the supported network to correct the state estimates. More precisely, each estimator running at the  $i$ th vehicle is described by

- For  $t_k^{[i]} \leq t < t_{k+1}^{[i]}$

$$\dot{\hat{\gamma}}_\gamma^{[i]} = f_\gamma^{[i]}(\hat{\gamma}_j^{[i]}, \cdot), \quad \forall \gamma \in \{i\} \cup \mathcal{N}_{t_k}^{[i]}$$

- For  $t = t_{k+1}^{[i]}$

$$\hat{\gamma}_\gamma^{[i]} = \gamma^{[\gamma]}, \quad \gamma \in \{i\} \cup \mathcal{N}_{t_{k+1}}^{[i]}$$

where the vector fields  $f_\gamma^{[i]}(\hat{\gamma}_j^{[i]}, \cdot)$  could be in general functions of the corresponding coordinated state estimate and other states. A simple choice is to make  $f_\gamma^{[i]}(\hat{\gamma}_j^{[i]}, \cdot) = 0$ , which means that (19) is just a piecewise continuous sample and hold discretization of (18). Other approaches include the work in [6] that shows how a logic-based communication can be incorporated in each of the vehicles in order to effectively decide when to transmit information to its neighbors, and the recent work described in [43] that proposes a self-triggered control strategy.

Note that the coordination is achieved by resorting to a decentralized control law whereby the exchange of data among the vehicles is kept at a minimum. In the consensus literature (see, e.g., [32], [36], [45]), there exist several algorithms and conditions under which the coordination is achieve, even when the topology of communication links among the vehicles is time-varying. For the simple neighbouring rule proposed, it is shown in [36] that consensus is achieved in the case where the communication graph is intermittently connected with brief connectivity losses, and where the union of the communication graphs over uniform intervals of time remains connected (uniformly connected in mean). Regarding the stability of the overall system, it is shown in [6], [36] that the system obtained by putting together the path-following and the vehicle coordination can be either a feedback interconnection or a cascade of two input-to-state stable (ISS) systems by assuming that individually each block is ISS. Stability and convergence properties of the resulting interconnected system can be then formally concluded through a small gain theorem based argument.

2) *Simulation results*: Consider three underactuated autonomous underwater vehicles. Vehicle 2 is allowed to communicate with vehicles 1 and 3, but the latter two do not communicate between themselves directly. To simulate losses in the communications, we considered the situation

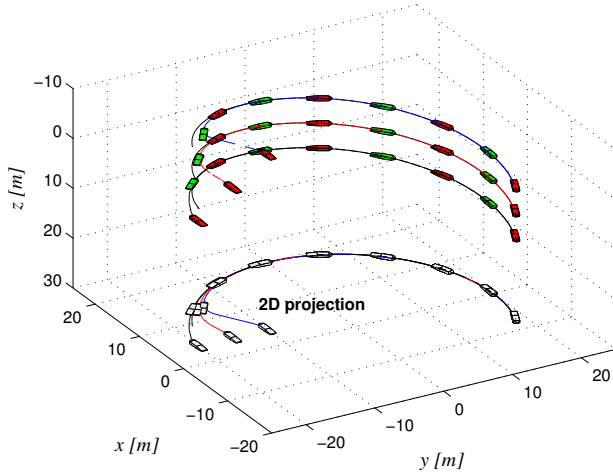


Fig. 9. Coordination of 3 AUVs, with communication failures.

where both links fail 75% of the time, with the failures occurring periodically with a period of 10 s. Notice that during failures all the links become deactivated.

In the simulations, the AUVs are required to follow three similar spatial paths shifted along the depth coordinate, that is, the paths are of the form

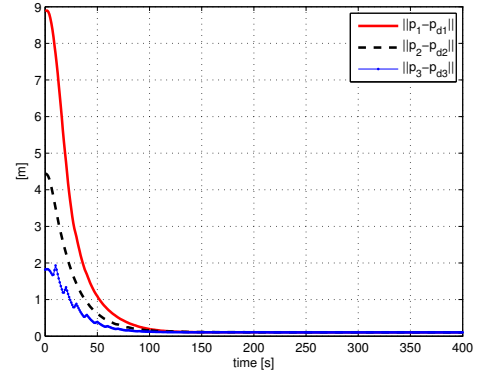
$$y_d^{[i]}(\gamma^{[i]}) = [c_1 \cos(\frac{2\pi}{T}\gamma^{[i]} + \phi_d), c_1 \sin(\frac{2\pi}{T}\gamma^{[i]} + \phi_d), c_2\gamma^{[i]} + z_{0_i}]$$

with  $c_1 = 20$  m,  $c_2 = 0.05$  m,  $T = 400$ ,  $\phi_d = -\frac{3\pi}{4}$ , and  $z_{0_1} = -10$  m,  $z_{0_2} = -5$  m,  $z_{0_3} = 0$  m. The initial configurations are  $p^{[1]} = (5 \text{ m}, -10 \text{ m}, -5 \text{ m})$ ,  $p^{[2]} = (5 \text{ m}, -15 \text{ m}, 0 \text{ m})$ ,  $p^{[3]} = (5 \text{ m}, -20 \text{ m}, 5 \text{ m})$ ,  $R^{[1]} = R^{[2]} = R^{[3]} = I$ , and  $\nu^{[1]} = \nu^{[2]} = \nu^{[3]} = \omega^{[1]} = \omega^{[2]} = \omega^{[3]} = 0$ .

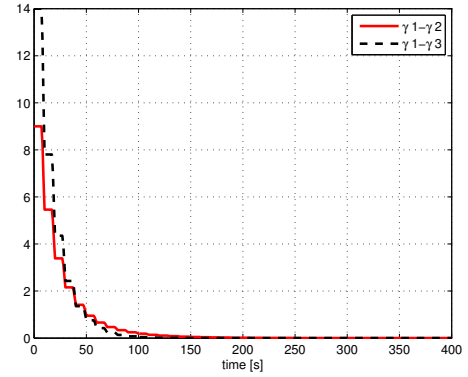
The vehicles are also required to keep a formation pattern that consists of having them aligned along a common vertical line. Fig. 9 shows the trajectories of the AUVs. Fig. 10 illustrates the evolution of the coordination and path-following errors when the communication links fail periodically. Clearly, the vehicles adjust their speeds to meet the formation requirements and the coordination errors  $\gamma^{[1]} - \gamma^{[2]}$  and  $\gamma^{[1]} - \gamma^{[3]}$  converge to zero.

#### IV. CONCLUSION

This paper presented a brief overview of selected motion control problems for single and multiple autonomous robotic vehicles. Many of the problems described here were motivated by challenging scientific mission scenarios defined in the course of several international research and development projects that the author and the co-authors of the papers referred have been participated. A general architecture for cooperative autonomous marine vehicle control in the presence of time-varying communication topologies and communication losses was highlighted. The architecture implementation relies on a number of single and multiple vehicle tasks, the development of which was rooted in solid control theory. The results of software and hardware-in-the-loop simulations, and



(a) Path-following errors



(b) Vehicle coordination errors

Fig. 10. 75% of temporal communication failures.

many field test series carried out in real scenarios have shown the efficacy of the methods described.

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#### REFERENCES

- [1] A. P. Aguiar, J. P. Hespanha, and P. V. Kokotovic. Path-following for nonminimum phase systems removes performance limitations. *IEEE Trans. on Automatic Control*, 50(2):234–239, feb 2005.
- [2] A. P. Aguiar, J. P. Hespanha, and P. V. Kokotović. Performance limitations in reference tracking and path following for nonlinear systems. *Automatica*, 44(3):598–610, mar 2008.
- [3] A. P. Aguiar, I. Kaminer, R. Ghabcheloo, A. M. Pascoal, E. Xargay, N. Hovakimyan, C. Cao, and V. Dobrokhodov. Coordinated path following of multiple UAVs for time-critical missions in the presence of time-varying communication topologies. In *Proc of the 17th IFAC World Congress, Seoul, Korea*, pages 16015–16020, jul 2008.
- [4] A. P. Aguiar and A. M. Pascoal. Regulation of a nonholonomic autonomous underwater vehicle with parametric modeling uncertainty using Lyapunov functions. In *Proc. 40th IEEE Conference on Decision and Control*, Orlando, Florida, USA, December 2001.
- [5] A. P. Aguiar and A. M. Pascoal. Global stabilization of an underactuated autonomous underwater vehicle via logic-based switching. In *Proc. 41st IEEE Conference on Decision and Control*, Las Vegas, NV, USA, December 2002.

- [6] A. P. Aguiar and A. M. Pascoal. Coordinated path-following control for nonlinear systems with logic-based communication. In *IEEE Conference on Decision and Control*, pages 1473–1479, 2007.
- [7] A. P. Aguiar, A. Rucco, and A. Alessandretti. A Sampled-Data Model-Predictive Framework for Cooperative Path Following of Multiple Robotic Vehicles. In *Lecture Notes in Control and Information Sciences*. 2017.
- [8] A. Pedro Aguiar and Joao P Hespanha. Trajectory-tracking and path-following of underactuated autonomous vehicles with parametric modeling uncertainty. *IEEE Transactions on Automatic Control*, 52(8):1362–1379, 2007.
- [9] A. Pedro Aguiar, João P Hespanha, and António M Pascoal. Switched seesaw control for the stabilization of underactuated vehicles. *Automatica*, 43(12):1997–2008, 2007.
- [10] A. Pedro Aguiar, Petar V Kokotovic, et al. Zero dynamics and tracking performance limits in nonlinear feedback systems. In *Analysis and Design of Nonlinear Control Systems*, pages 149–159. Springer, 2008.
- [11] A. Pedro Aguiar and António M Pascoal. Dynamic positioning and way-point tracking of underactuated auvs in the presence of ocean currents. *International Journal of Control*, 80(7):1092–1108, 2007.
- [12] M. Aicardi, G. Casalino, A. Bicchi, and A. Balestrino. Closed loop steering of unicycle-like vehicles via Lyapunov techniques. *IEEE Robotics & Automation Magazine*, 2(1):27–35, 1995.
- [13] Andrea Alessandretti, A. Pedro Aguiar, and Colin N Jones. Trajectory-tracking and path-following controllers for constrained underactuated vehicles using model predictive control. In *Control Conference (ECC), 2013 European*, pages 1371–1376. IEEE, 2013.
- [14] Andrea Alessandretti, A. Pedro Aguiar, and Colin N Jones. An input-to-state-stability approach to economic optimization in model predictive control. *IEEE Transactions on Automatic Control*, 2017.
- [15] A. Astolfi. Discontinuous control of the brockett integrator. *European Journal of Control*, 4(1):49–63, 1998.
- [16] R. Beard, J. Lawton, and F. Hadaegh. A coordination architecture for spacecraft formation control. *IEEE Trans. on Control Systems Technology*, 9:777–790, 2001.
- [17] L. Bishop. There is more than one way to frame a curve. In *Amer. Math.*, pages 246–251. Monthly 82, March 1975.
- [18] A. Bloch and S. Drakunov. Stabilization of a nonholonomic system via sliding modes. In *Proc. 33rd IEEE Conference on Decision and Control*, Orlando, Florida, USA, December 1994.
- [19] A. M. Bloch, M. Reyhanoglu, and N. H. McClamroch. Control and stabilization of nonholonomic dynamic systems. *IEEE Transactions on Automatic Control*, 37(11):1746–1757, 1992.
- [20] Even Børhaug, Alexey Pavlov, Elena Panteley, and Kristin Y Pettersen. Straight line path following for formations of underactuated marine surface vessels. *IEEE Transactions on Control Systems Technology*, 19(3):493–506, 2011.
- [21] Morten Breivik, Vegard E Hovstein, and Thor I Fossen. Straight-line target tracking for unmanned surface vehicles. *Modeling, Identification and Control*, 29(4):131–149, 2008.
- [22] Roger W. Brockett. Asymptotic stability and feedback stabilization. *Differential Geometric Control Theory*, (Defense Technical Information Center):181–191, 1983.
- [23] C. Canudas de Wit, H. Khenouf, Claude Samson, and O Sordalen. Nonlinear control design for mobile robots. *Recent Trends in Mobile Robots*, 11:121–156, 1993.
- [24] C. Canudas de Wit and O.J. Sordalen. Exponential stabilization of mobile robots with nonholonomic constraints. *IEEE Transactions on Automatic Control*, 37(11):1791–1797, 1992.
- [25] Venanzio Cichella, Ronald Choe, Syed Bilal Mehdi, Enric Xargay, Naira Hovakimyan, Vladimir Dobrokhodov, Isaac Kaminer, Antonio M Pascoal, and A. Pedro Aguiar. Safe coordinated maneuvering of teams of multirotor unmanned aerial vehicles: A cooperative control framework for multivehicle, time-critical missions. *IEEE Control Systems*, 36(4):59–82, 2016.
- [26] Venanzio Cichella, Isaac Kaminer, Vladimir Dobrokhodov, Enric Xargay, Ronald Choe, Naira Hovakimyan, A. Pedro Aguiar, and António M Pascoal. Cooperative path following of multiple multirotors over time-varying networks. *IEEE Transactions on Automation Science and Engineering*, 12(3):945–957, 2015.
- [27] J. Desai, J. Ostrowski, and V. Kumar. Controlling formations of multiple mobile robots. In *Proceedings IEEE International Conference on Robotics and Automation*, volume 4, pages 2864–2869, 1998.
- [28] H. Dias, P. Calado, R. Bencatel, R. Gomes, S. Ferreira, and J. Sousa. Operations with multiple unmanned systems. In *International Conference on Intelligent Robots and Systems (IROS)*, 2012.
- [29] W. Dixon, Z. Jiang, and D. Dawson. Global exponential setpoint control of wheeled mobile robots: a lyapunov approach. *Automatica*, 36:1741–1746, 2000.
- [30] M. Egerstedt and X. Hu. Formation constrained multi-agent control. *IEEE Transactions on Robotics and Automation*, 17(6):947–951, December 2001.
- [31] Pedro Encarnao and Antnio Pascoal. 3D path following for autonomous underwater vehicles. In *Proceedings of the 39th Conference on Decision and Control*, pages 2977 – 2982, 2000.
- [32] J. A. Fax and R. M. Murray. Graph laplacians and stabilization of vehicle formations. In *IFAC World Congress*, Barcelona, Spain, 2002.
- [33] Thor I Fossen, Morten Breivik, and Roger Skjetne. Line-of-sight path following of underactuated marine craft. *Proceedings of the 6th IFAC MCMC, Girona, Spain*, pages 244–249, 2003.
- [34] E. Frew, D. Lawrence, C. Dixon, J. Elston, and J. Pisano. Lyapunov guidance vector fields for unmanned aircraft applications. In *Proceedings of the 2007 American Control Conference*, pages 371–376, 2007.
- [35] R. Ghabcheloo, A. P. Aguiar, A. Pascoal, C. Silvestre, I. Kaminer, and J. Hespanha. Coordinated path-following control of multiple underactuated autonomous vehicles in the presence of communication failures. *Proc 45th IEEE Conference on Decision and Control (CDC), San Diego, California, USA*, pages 4345 – 4350, 2006.
- [36] R. Ghabcheloo, A. P. Aguiar, A. M. Pascoal, C. Silvestre, Isaac Kaminer, and J. P. Hespanha. Coordinated path-following in the presence of communication losses and time delays. *SIAM Journal on Control and Optimization*, 48(1):234–265, 2009.
- [37] R. Ghabcheloo, A. Pascoal, C. Silvestre, and I. Kaminer. Nonlinear coordinated path following control with bidirectional communication constraints. In J. Gravidahl K. Pettersen and H. Nijmeijer, editors, *Group Coordination and Cooperative Control*, volume 336, pages 93–111. Springer Series on Lecture Notes in Control and Information Sciences, 2006.
- [38] F. Giuletti, L. Pollini, and M. Innocenti. Autonomous formation flight. *IEEE Control Systems Magazine*, 20(6):34–44, 2000.
- [39] J. M. Godhavn and O. Egeland. A Lyapunov approach to exponential stabilization of nonholonomic systems in power form. *IEEE Transactions on Automatic Control*, 42(7):1028–1032, 1997.
- [40] J. P. Hespanha. Stabilization of nonholonomic integrators via logic-based switching. In *Proc. 13th World Congress of IFAC*, volume E, pages 467–472, S. Francisco, CA, USA, June 1996.
- [41] J. P. Hespanha, Payam Naghshtabrizi, and Yonggang Xu. A survey of recent results in networked control systems. *Proc of IEEE Special Issue on Technology of Networked Control Systems*, 95(1):138–162, 2007.
- [42] Ivar-André F Ihle, Murat Arcak, and Thor I Fossen. Passivity-based designs for synchronized path-following. *Automatica*, 43(9):1508–1518, 2007.
- [43] R. Praveen Kumar Jain, A. Pedro Aguiar, and Joao Borges Sousa. Self-triggered cooperative path following control of fixed wing unmanned aerial vehicles. In *2017 International Conference on Unmanned Aircraft Systems (ICUAS)*, pages 1231–1240. IEEE, 2017.
- [44] I. Kaminer, A. Pascoal, C. Cao, and V. Dobrokhodov. Path following for unmanned aerial vehicles using L1 adaptive augmentation of commercial autopilots. *Journal of Guidance, Control and Dynamics*, 33:550–564, 2010.
- [45] I. Kaminer, E. Xargay, V. Cichella, N. Hovakimyan, A. M. Pascoal, A. P. Aguiar, V. Dobrokhodov, and R. Ghabcheloo. Time-critical cooperative path following of multiple UAVs: Case studies. In *Advances in Estimation, Navigation, and Spacecraft Control*, pages 209–233. Springer, 2015.
- [46] Isaac Kaminer, O. Yakimenko, Antnio Pascoal, and Reza Ghabcheloo. Path generation, path following and coordinated control for time-critical missions of multiple UAVs. In *Proceedings of the 2006 American Control Conference*, pages 4906–4913, 2006.
- [47] H. Khalil. *Nonlinear Systems*, 3 edition, 2002. Prentice Hall.
- [48] L. Lapierre, D. Soetanto, and A. Pascoal. Adaptive, non-singular path-following control of dynamic wheeled robots. In *Proceedings of the 42nd IEEE Conference on Decision and Control*, vol. 2, pages 1765–1770, 2003.
- [49] David A. Lizárraga, Nnaedozie P. I. Aneke, and Henk Nijmeijer. Robust point stabilization of underactuated mechanical systems via the extended chained form. *SIAM J. Contr. Optimization*, 42(6):2172–2199, 2004.

- [50] David B Marco and Anthony J Healey. Command, control, and navigation experimental results with the nps aries auv. *IEEE Journal of Oceanic Engineering*, 26(4):466–476, 2001.
- [51] R. T. M'Closkey and R. M. Murray. Exponential stabilization of driftless nonlinear control systems using homogeneous feedback. *IEEE Transactions on Automatic Control*, 42(5):614–628, 1997.
- [52] P. Morin and C. Samson. Control of nonlinear chained systems: From the Routh-Hurwitz stability criterion to time-varying exponential stabilizers. *IEEE Transactions on Automatic Control*, 45(1):141–146, 2000.
- [53] Pascal Morin and Claude Samson. Practical stabilization of driftless systems on lie groups: The transverse function approach. *IEEE Trans. on Automat. Contr.*, 48(9):1496–1508, September 2003.
- [54] P. Ogren, E. Fiorelli, and N. Leonard. Cooperative control of mobile sensor networks: Adaptive gradient climbing in a distributed environment. *IEEE Trans on Automatic Control*, 49(8):1292–1302, 2004.
- [55] T Oliveira and P Encarnao. Ground target tracking control system for unmanned aerial vehicles. *Journal of Intelligent and Robotic Systems*, 69:373–387, 2013.
- [56] Tiago Oliveira, A. Pedro Aguiar, and Pedro Encarnacao. Moving Path Following for Unmanned Aerial Vehicles With Applications to Single and Multiple Target Tracking Problems. *IEEE Transactions on Robotics*, 32(5):1062–1078, oct 2016.
- [57] Tiago Oliveira, A. Pedro Aguiar, and Pedro Encarnacao. Three dimensional moving path following for fixed-wing unmanned aircraft vehicles. In *Proceedings 2017 IEEE International Conference on Robotics and Automation, 2017. ICRA 2017*, May 2017.
- [58] A. Pascoal, C. Silvestre, and P. Oliveira. Vehicle and mission control of single and multiple autonomous marine robots. In G.N. Roberts and R. Sutton, editors, *Advances in Unmanned Marine Vehicles, IEE Control Engineering Series, G*, volume 69. IET, 2006. Chapter 17.
- [59] Christophe Prieur and Alessandro Astolfi. Robust stabilization of chained systems via hybrid control. *IEEE Trans. on Automat. Contr.*, 48(10):1768–1772, October 2003.
- [60] Alessandro Rucco, A. Pedro Aguiar, and John Hauser. Trajectory optimization for constrained UAVs: A Virtual Target Vehicle approach. In *2015 International Conference on Unmanned Aircraft Systems (ICUAS)*, pages 236–245. IEEE, jun 2015.
- [61] Alessandro Rucco, António Pedro Aguiar, Fernando ACC Fontes, Fernando Lobo Pereira, and João Borges de Sousa. A model predictive control-based architecture for cooperative path-following of multiple unmanned aerial vehicles. In *Developments in Model-Based Optimization and Control*, pages 141–160. Springer, 2015.
- [62] Mauro Salazar, Andrea Alessandretti, A Pedro Aguiar, and Colin N Jones. An energy efficient trajectory tracking controller for car-like vehicles using model predictive control. In *Decision and Control (CDC), 2015 IEEE 54th Annual Conference on*, pages 3675–3680. IEEE, 2015.
- [63] C. Samson. Control of chained systems: Application to path following and time-varying point-stabilization of mobile robots. *IEEE Transactions on Automatic Control*, 40(1):64–77, 1995.
- [64] Claude Samson. Path following and time-varying feedback stabilization of wheeled mobile robots. In *Proceedings of the International Conference on Advanced Robotics and Computer Vision*, volume 13, 1992.
- [65] R. Skjetne, I. Flakstad, and T. Fossen. Formation control by synchronizing multiple maneuvering systems. *Proc 6th IFAC Conference on Maneuvering and Control of Marine Craft, Girona, Spain*, 2003.
- [66] R. Skjetne, T. Fossen, and P. Kokotovic. Robust output maneuvering for a Class of nonlinear systems. *Automatica*, 40(3):373–383, 2004.
- [67] R. Skjetne, S. Moi, and T. I. Fossen. Nonlinear formation control of marine craft. In *Proceedings of the 41<sup>st</sup> IEEE Conference on Decision and Control (CDC'02)*, volume 2, pages 1699–1704, Las Vegas, NV, USA, December 2002.
- [68] A. R. Tell, R. M. Murray, and G. C. Walsh. Nonholonomic control systems: from steering to stabilization with sinusoids. *International Journal of Control*, 62:849–870, 1995.
- [69] Richard Wise and Rolf Rysdyk. UAV coordination for autonomous target tracking. In *Proceedings of the AIAA Guidance, Navigation, and Control Conference and Exhibit*, 2006.
- [70] E Xargay, I Kaminer, A Pascoal, N Hovakimyan, V Dobrokhodov, V Cichella, AP Aguiar, and R Ghabcheloo. Time-critical cooperative path following of multiple unmanned aerial vehicles over time-varying networks. *Journal of Guidance, Control, and Dynamics*, 36(2):499–516, 2013.
- [71] J. K. Yook, D. M. Tilbury, and N. R. Soparkar. Trading computation for bandwidth: Reducing communication in distributed control systems using state estimators. *IEEE Transactions on Control Systems Technology*, 10(4):503–518, July 2002.
- [72] Mingfeng Zhang and H Liu. Game-theoretical persistent tracking of a moving target using a unicycle-type mobile vehicle. *IEEE Transactions on Industrial Electronics*, 61:6222–6233, 2014.